

female-to-female jumper wires.

To configure XBee module, connect XBee Explorer shield to a computer via USB.

Then run the XCTU software and click on 'Add Devices' icon. The

output screenshot of XCTU is shown in Fig. 2. You will get an 'Add radio device' window. Confirm the COM port of the Explorer and set the baud rate to 9600, data bits to 8, parity to none and click on 'Finish'

to the computer with XCTU should be set as end device. Router (sort of repeater) is an optional mode; in this, XBee simply receives and then transmits the same data, and so this mode is used to extend the range of transmitter.

Now first, configure XBee as coordinator. To do that, click on 'Update Firmware,' set it to 'XBee coordinator AT,' click on the latest firmware available and then click on 'Finish.' Each XBee has two important keys, SH and SL, which help in transmission and reception of data. SL is unique to each XBee whereas SH is the same. The bottom view of a typical XBee module is shown in Fig. 5 along with its SL number as 406F70BE and SH as 0013A200.

Let us consider an XBee module (XB1) as the coordinator with its SL number as 40D6A4AB and SH as 0013A200. Now consider another XBee module (XB2), which is used as receiver or end device, with its SL number as 40D86AE4 and SH number the same, that is, 0013A200. In Radio Configuration Window, enter the DL and DH values as the SL and SH values of XB2 and click on 'Write Radio Settings.'

To configure XB2 as end device, repeat the steps done for configuring XB1, update the firmware and replace its DH and DL values with the SH and SL values of coordinator XB1, respectively. Now XB1 can transmit data and XB2 can receive it. You can see the details of coordinator (XB1) and end device (XB2) modules in Fig. 6 and Fig. 7, respectively.

Interfacing XBee with Raspberry Pi

The communication between XBee and Raspberry Pi is done through serial communication. The connections for serial communication between XBee and Raspberry Pi Model 2 are given below and also shown in Fig. 8.

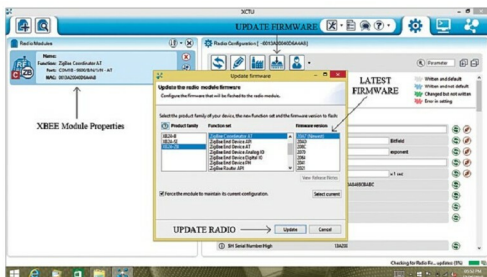


Fig. 4: Radio Configuration window



Fig. 5: Bottom view of a typical XBee module

as shown in Fig. 3.

Now click on 'XBee Module' on the left of the Radio Configuration screen (refer Fig. 4) to configure it. There are three types of devices in a communication network: coordinator, end device and router. The coordinator sends the data, so the XBee connected to Raspberry Pi is set as coordinator. End device receives the data. Here we will use Console Mode of X-CTU to view received data, hence, XBee connected

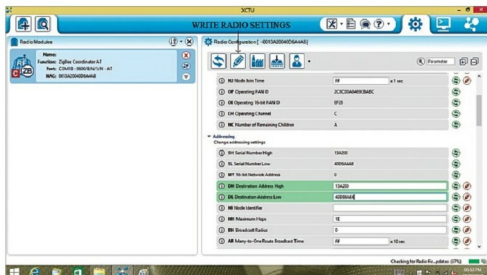


Fig. 6: Configuration details of coordinator